

Technical Specifications
Packaging Machine: HANNA

size:	150x420mm
fold:	150x35mm
carton weight:	60gsm
standard number of colors	1

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PRODUCT NAME

Merazole (Letrozole Film Coated Tablets 2.5mg)

NAME AND STRENGTH OF ACTIVE INGREDIENT

Merazole 2.5mg

PRODUCT DESCRIPTION

Yellow, round, biconvex film-coated tablets

PHARMACODYNAMICS

In healthy postmenopausal women, single doses of 0.1 mg, 0.5 mg and 2.5 mg letrozole suppress serum estrone and estradiol by 75 to 78% and 78% from baseline, respectively. Maximum suppression is achieved in 48 to 72 hours.

In postmenopausal patients with advanced breast cancer daily doses of 0.1 to 5 mg suppress plasma concentration of estradiol, estrone, and estrone sulphate by 75 to 95% from baseline in all patients treated. With doses of 0.5 mg and higher, many values of estrone and estrone sulphate are below the limit of detection in the assays, indicating that higher estrogen suppression is achieved with these doses. Estrone suppression was maintained throughout treatment in all these patients.

Letrozole is highly specific in inhibiting aromatase activity. Impairment of adrenal steroidogenesis has not been observed. No clinically relevant changes were found in the plasma concentrations of cortisol, aldosterone, 11-deoxycortisol, 17-hydroxy-progesterone, and ACTH, or in plasma renin activity among postmenopausal patients treated with a daily dose of letrozole 0.1 to 5 mg. The ACTH stimulation test performed after 6 and 12 weeks of treatment with daily doses of 0.1 mg, 0.25 mg, 0.5 mg, 1 mg, 2.5 mg, and 5 mg did not indicate any alteration of adrenal or cortisol production. Thus, glucocorticoid and mineralocorticoid supplementation is not necessary.

No changes were noted in plasma concentrations of androgens (androstenedione and testosterone) among healthy postmenopausal women after 0.1 mg, 0.5 mg, and 2.5 mg single doses of letrozole or in plasma concentrations of androstenedione among postmenopausal patients treated with daily doses of 0.1 to 5 mg, indicating that the blockade of estrogen biosynthesis does not lead to accumulation of androgenic precursors. Plasma levels of LH and FSH are not affected by letrozole in patients, nor is thyroid function as evaluated by TSH, T4 and T3 uptake.

PHARMACOKINETICS

Absorption
Letrozole is rapidly and completely absorbed from the gastrointestinal tract (mean absolute bioavailability: 593%). Food slightly decreases the rate of absorption (median t_{max} 1 hour; fasted versus 2 hours; fed) and mean C_{max} 120 ± 203 nmol/L (fasted versus 98.7 ± 18.6 nmol/L, fed), but the extent of absorption (AUC) is not changed. The minor effect on the absorption rate is not considered to be of clinical relevance, and therefore letrozole may be taken without regard to meal times.

Distribution
Plasma protein binding of letrozole is approximately 60%, mainly to albumin (55%). The concentration of letrozole in erythrocytes is about 40% of that in plasma. After administration of 2.5 mg ^{14}C -labelled letrozole, approximately 82% of the radioactivity in plasma was unchanged compound. Systemic exposure to metabolites is therefore low. Letrozole is rapidly and extensively distributed to tissues. Its apparent volume of distribution at steady state is about 187 ± 0.47 L/kg.

Biotransformation/metabolism
Metabolic clearance to a pharmacologically inactive carbonyl metabolite is the major elimination pathway of letrozole (CL_{int} = 2.1 L/h), but is relatively slow when compared to hepatic blood flow (about 90 L/h). The cytochrome P₄₅₀ isoenzymes 3A4 and 2A6 were found to be capable of converting letrozole to this metabolite. Formation of minor unidentified metabolites, and direct renal and faecal excretion play only a minor role in the overall elimination of letrozole. Within 2 weeks after administration of 2.5 mg ^{14}C -labelled letrozole to healthy postmenopausal volunteers 88.2 ± 7.6% of the radioactivity was recovered in urine and 3.8 ± 0.9 % in faeces. At least 75 % of the radioactivity recovered in urine up to 216 hours (84.7 ± 7.8% of the dose) was attributed to the glucuronide of the carbonyl metabolite, about 9 % to two unidentified metabolites, and 6 % to unchanged letrozole.

Elimination
The apparent terminal elimination half-life in plasma is between about 2 to 4 days. After daily administration of 2.5 mg, steady-state levels are reached within 2 to 6 weeks. Plasma concentrations at steady state are approximately 2 times higher than concentrations measured after a single dose of 2.5 mg, while they are 1.5 to 2 times higher than the steady-state values predicted from the concentrations measured after a single dose, indicating a slight non-linearity in the pharmacokinetics of letrozole upon daily administration of 2.5 mg. Since steady-state levels are maintained over time, it can be concluded that no continuous accumulation of letrozole occurs.

Linearity/non-linearity

The pharmacokinetics of letrozole were dose proportional after single oral doses up to 10 mg (dose range: 0.03 to 30 mg) and after daily doses up to 10 mg (dose range: 0.03 to 5 mg). After a 30 mg single oral dose there was a slightly dose over-proportional increase in AUC value. With daily doses of 2.5 and 5 mg the AUC values increased about 3.8 and 12 fold instead of 2.5 and 5 fold, respectively, when compared to the 1.0 mg/day dose. The recommended dose of 2.5 mg/day may thus be a borderline dose at which an onset of over-proportionality becomes apparent, whereas at 5 mg/day the over-proportionality is more pronounced. The dose over-proportionality is likely to be the result of a saturation of metabolic elimination processes. Steady levels were reached after 1 to 2 months at all dosage regimens tested (0.1-50 mg/day).

Special Populations

Elderly
Age had no effect on the pharmacokinetics of letrozole.

Renal impairment

In a study involving 16 postmenopausal volunteers with varying degrees of renal function (24-hour creatinine clearance 9 to 116 mL/min), no effect on the pharmacokinetics systemic exposure of letrozole was found after a single dose of 2.5 mg. In addition to the above study assessing the influence of renal impairment on letrozole, a covariate analysis was performed on the data of two pivotal studies (Study AR/B/C2 and Study AR/B/C3). Calculated creatinine clearance (CL_{Cr}) [Study AR/B/C2 range 10 to 187 mL/min; Study AR/B/C3 range 10 to 180 mL/min] showed no statistically significant association between letrozole plasma trough levels at steady-state (C_{min}). Furthermore, data of Study AR/B/C2 and Study AR/B/C3 in second-line metastatic breast cancer showed no evidence of an adverse effect of letrozole on CL_{Cr} or an impairment of renal function.

Therefore, no dose adjustment is required for patients with renal impairment (CL_{Cr} ≥ 40 mL/min). Little information is available in patients with severe impairment of renal function (CL_{Cr} < 10 mL/min).

Hepatic impairment

In a similar study involving subjects with varying degrees of hepatic function, the mean AUC values of the volunteers with moderate hepatic impairment (Child-Pugh score B) was 37 % higher than in normal subjects, but still within the range seen in subjects without impaired function. In a study comparing the pharmacokinetics of letrozole after a single oral dose in eight subjects with liver cirrhosis and severe hepatic impairment (Child-Pugh score C) to those in healthy volunteers (n=8), AUC and $t_{1/2}$ increased by 5 and 187%, respectively. Breast-cancer patients with severe hepatic impairment are thus expected to be exposed to higher levels of letrozole than patients without severe hepatic dysfunction. However, since in patients dosed at 5 or 10 mg/day no increase in toxicity was observed, a dose reduction in patients with severe hepatic impairment appears not to be warranted, although such patients should be kept under close supervision. In addition, in two well-controlled studies involving 259 patients with advanced breast cancer, no effect of renal impairment (calculated creatinine clearance: 20 to 50 mL/min) or hepatic dysfunction was found on the letrozole concentration.

INDICATION

- Adjuvant treatment of postmenopausal women with hormone receptor positive-early breast cancer.
- Adjuvant treatment of postmenopausal women with early breast cancer (positive or unknown oestrogen or progesterone receptor status) who have received 5 years of adjuvant tamoxifen therapy (extended adjuvant therapy).
- First-line treatment in postmenopausal women with hormone-dependent advanced breast cancer.
- In patients of advanced breast cancer in women with natural or artificially induced postmenopausal status, who have previously been treated with antiestrogens.
- The operative therapy in postmenopausal women with localised hormone receptor positive breast cancer, to reduce subsequent breast-conserving surgery in women not originally considered candidates for this type of surgery. Subsequent treatment after surgery should be in accordance with standard of care.

RECOMMENDED DOSAGE

General Target Population

Adults

The recommended dose of Merazole is 2.5 mg once daily in the morning with food. The recommended treatment with Merazole should continue for 5 years or until disease relapse/recurrence occurs, whichever comes first. In the large pivotal study of letrozole versus tamoxifen in the adjuvant setting, no benefit in efficacy or safety was observed by sequential administration of these treatments compared with letrozole administered continuously for 5 years. In patients with metastatic disease, treatment with Merazole should continue until tumor progression is evident. In the pre-operative setting, treatment with Merazole should be continued for 4 to 8 weeks prior to establish optimal tumor reduction. If the response is not adequate, treatment with Merazole should be discontinued, surgery scheduled and/or further treatment options discussed with the patient.

Special Populations

Hepatic impairment

No dose adjustment of Merazole is required for patients with mild to moderate hepatic insufficiency (Child-Pugh score A or B). Insufficient data are available for patients with severe hepatic impairment, but patients with severe hepatic impairment (Child-Pugh score C) should be kept under close supervision.

Renal impairment

No dosage adjustment of Merazole is required for patients with renal insufficiency with creatinine clearance (CL_{Cr} ≥ 10 mL/min). Insufficient data are available in cases of renal insufficiency with CL_{Cr} < 10 mL/min.

Pediatrics

Merazole is not recommended for use in children and adolescents. The safety and efficacy of Merazole in children and adolescents aged up to 17 years have not been established. Limited data are available and no clinical studies in this age population can be made.

Geriatrics

No dose adjustment is required for elderly patients.

Method of Administration

Merazole should be taken orally and can be taken with or without food because food has no effect on the extent of absorption.

Missed Dose

The missed dose should be taken as soon as the patient remembers to take it. If it is almost time for the next dose, the missed dose should be skipped, and the patient should go back to her regular dosage schedule. Doses should not be doubled because with daily doses over the 2.5 mg recommended dose, over-proportionality in systemic exposure was observed.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

- Known hypersensitivity to the active substance or to any of the excipients.
- Premenopausal endocrine status, pregnancy, lactation.

WARNING AND PRECAUTIONS

Renal impairment

Merazole has not been investigated in patients with creatinine clearance < 10 mL/min. The potential risk/benefit to such patients should be carefully considered before administration of Merazole.

Hepatic impairment

In patients with severe hepatic impairment (Child-Pugh score C), systemic exposure and terminal half-life were approximately doubled compared to healthy volunteers. Such patients should therefore be kept under close supervision.

Bone Effects

Osteoporosis and/or bone fractures have been reported with the use of Merazole. Therefore, monitoring of overall bone health is recommended during treatment.

Menopausal Status

In patients whose menopausal status is unclear, luteinizing hormone (LH), follicle-stimulating hormone (FSH) and/or estradiol levels should be measured before initiating treatment with Merazole. Only women of confirmed postmenopausal endocrine status should receive Merazole.

Fertility

The pharmacological action of letrozole is to reduce estrogen production by aromatase inhibition. In premenopausal women, the inhibition of estrogen synthesis leads to feedback increases in gonadotropin (LH, FSH) levels. Increased FSH levels in turn stimulate follicular growth and can induce ovulation.

Interactions

Co-administration of Merazole with tamoxifen, other anti-estrogens or estrogen-containing therapies should be avoided as these substances may diminish the pharmacological action of letrozole. The mechanism of this interaction is unknown.

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INTERACTIONS WITH OTHER MEDICATIONS

Letrozole is mainly metabolized in the liver and the cytochrome P450 enzymes CYP3A4 and CYP2A6 mediate the metabolic clearance of letrozole. Therefore, the systemic elimination of letrozole may be influenced by drugs known to affect the CYP3A4 and CYP2A6. The metabolism of letrozole appears to have a low affinity for CYP3A4 because the enzyme could not be saturated at concentrations more than 150 fold higher than those observed in plasma for letrozole at steady-state in the typical clinical situation.

Drugs that may increase Letrozole serum concentrations
Inhibitors of CYP3A4 and CYP2A6 activities could decrease the metabolism of letrozole and thereby increase plasma concentrations of letrozole. The concomitant administration of medications that strongly inhibit these enzymes (strong CYP3A4 inhibitors) including but not limited to ketoconazole, itraconazole, voriconazole, itraconazole, clarithromycin, and telithromycin, and CYP2A6 (e.g., methoxsalen) may increase exposure to letrozole. Therefore, caution is recommended in patients for whom strong CYP3A4 and CYP2A6 inhibitors are administered.

Drugs that may decrease Letrozole serum concentrations
Inducers of CYP3A4 activity could increase the metabolism of letrozole and thereby decrease plasma concentrations of letrozole. The concomitant administration of medications that induce CYP3A4 (e.g., phenytoin, rifampicin, carbamazepine, phenobarbital, and St. John's Wort) may reduce exposure to letrozole. Therefore, caution is recommended in patients for whom strong CYP3A4 inducers are administered. No drug inducer is known for CYP2A6.

Co-administration of Merazole (2.5 mg) and tamoxifen 20 mg daily resulted in a reduction of letrozole plasma levels by 38 % on average. Clinical experience in the second-line breast cancer trials indicates that neither the therapeutic effect of Merazole therapy nor the occurrence of adverse reactions is increased if Merazole is administered immediately after tamoxifen. The mechanism of this interaction is unknown.

Drugs that may have their systemic serum concentrations altered by Letrozole

In vitro, letrozole inhibits the cytochrome P450 isoenzymes CYP2A6 and, moderately, CYP2C19, but the clinical relevance is unknown. Caution is therefore indicated when giving letrozole concomitantly with medicinal products whose elimination is mainly dependent on CYP2C19 and whose therapeutic index is narrow (e.g., phenytoin, clobazepam, clobazepam). No substrate with a narrow therapeutic index is known for CYP2A6.

Clinical interaction studies with cimetidine (a known non-specific inhibitor of CYP2C19 and CYP3A4 and warfarin (sensitive substrate for CYP2C9 with a narrow therapeutic window and commonly used as a co-medication in the target population of letrozole) indicated that the coadministration of Merazole with these drugs does not result in clinically significant drug interactions.

A review of the clinical trial database indicated no evidence of other clinically relevant interaction with other commonly prescribed drugs.

PREGNANCY AND LACTATION

Merazole is contraindicated during pregnancy.

There has been post-marketing reports of spontaneous abortions and congenital anomalies in infants of mothers who took Merazole during pregnancy.

Isolated cases of birth defects (labial fusion, ambiguous genitalia) have been reported in infants born to pregnant women exposed to off label use (infertility treatment, ovulation induction) of Merazole.

Merazole is contraindicated during lactation.

Females and males of reproductive potential

The physician needs to discuss the necessity of adequate contraception with women who have the potential to become pregnant. In including women who are perimenopausal or who recently became postmenopausal, until their postmenopausal status is fully established.

SIDE EFFECTS

Summary of Safety Profile

Merazole was generally well tolerated across all studies as first-line and second-line treatment for advanced breast cancer, as adjuvant treatment of early breast cancer and as extended adjuvant treatment in women who have received prior standard adjuvant therapy with tamoxifen. Approximately one third of the patients treated with Merazole in the metastatic and neoadjuvant settings, approximately 81% of the patients in the adjuvant setting (in both Merazole and tamoxifen arms), at a median treatment duration of 60 months, and approximately 80 % of the patients in the extended adjuvant setting (both Merazole and placebo arms, at a median treatment duration of 60 months) experienced adverse reactions. Generally, the observed adverse reactions are mild or moderate in nature, and many are associated with estrogen deprivation.

The most frequently reported adverse reactions in the clinical studies were hot flashes, arthralgia, nausea and fatigue. Many adverse reactions can be attributed to the normal pharmacological consequences of estrogen deprivation (e.g., hot flashes, alopecia and vaginal bleeding). The following adverse drug reactions, listed in Table 1, were reported from clinical studies and from post marketing experience with Merazole.

Table 1: Adverse drug reactions

Infections and infestations	Urinary tract infection
Uncommon	
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Tumor pain
Uncommon	
Blood and the lymphatic system disorders	Leukopenia
Uncommon	
Immune system disorders	Anaphylactic reaction
Unknown	
Metabolism and nutrition disorders	Hypercholesterolemia
Very common	Decreased appetite, increased appetite
Common	
Psychiatric disorders	Depression
Common	Anxiety (including nervousness), irritability
Uncommon	

Nervous system disorders	Headache, dizziness, vertigo
Common	Somnolence, insomnia, memory impairment, dysaesthesia (including paraesthesia, hypoesthesia), dysgeusia, cerebrovascular accident, carpal tunnel syndrome
Uncommon	
Eye disorders	Cataract, eye irritation, blurred vision
Uncommon	
Cardiac disorders	Palpitations
Common	Tachycardia ischaemic cardiac events (including new or worsening angina, angina requiring surgery, myocardial infarction and myocardial ischemia)
Uncommon	
Vascular disorders	Hot flush
Very common	Hypertension
Uncommon	Thrombophlebitis (including superficial and deep vein thrombosis)
Rare	Pulmonary embolism, arterial thrombosis, cerebral infarction
Respiratory, thoracic, and mediastinal disorders	Dyspnoea, cough
Common	
Gastrointestinal disorders	Nausea, vomiting, dyspepsia, constipation, diarrhoea, abdominal pain
Common	Stomatitis, dry mouth
Uncommon	
Hepatobiliary disorders	Increased hepatic enzymes, hyperbilirubinaemia, jaundice
Uncommon	Hepatitis
Very rare	
Skin and subcutaneous tissue disorders	Hyperhidrosis
Very common	Alopecia, dry skin, rash (including erythematous, maculopapular, psoriasiform and vesicular rash)
Common	Pruritus, urticaria
Uncommon	Angioedema, toxic epidermal necrolysis, erythema multiforme
Not known	
Musculoskeletal and connective tissue disorders	Arthralgia
Very common	Myalgia, bone pain, osteoporosis, bone fractures, arthritis, back pain
Common	Trigger finger
Not known	
Renal and urinary disorders	Polakiuria
Uncommon	
Reproductive system and breast disorders	Vaginal haemorrhage
Common	Abnormal discharge, vulvovaginal dryness, breast pain
Common	
General disorders and administration site conditions	Fatigue (including asthenia, malaise)
Very common	Peripheral oedema, chest pain
Common	General oedema, pyrexia, mucosal dryness, thirst
Uncommon	
Investigations	Weight increased
Common	Weight decreased
Uncommon	
Injury, poisoning and procedural complications	
Common	Fall
Common	

Adverse drug reactions reported only in the metastatic setting.
Frequency determined based on FACCE study data.
In some cases, follow-up was reported as a consequence of other adverse events such as dizziness and vertigo.

OVERDOSAGE AND TREATMENT
Isolated cases of overdosage with Merazole have been reported. No specific treatment for overdosage is known; treatment should be symptomatic and supportive.

EFFECTS ON ABILITY TO DRIVE AND OPERATE MACHINERY
Since fatigue and dizziness have been observed with the use of Merazole and somnolence has been reported uncommonly, caution is advised when driving or using machines.

STORAGE
Do not store above 30°C. Protect from moisture. Merazole should not be used after the date marked "EXP" on the pack.

DOSAGE FORM

Film Coated Tablets

PACKAGING AVAILABLE

Box of 3 x 10 tablets
10 tablets per PVC/PE/PVC/Aluminum blister

PRODUCT OWNER/MANUFACTURER

GenePharm S.A.
87 km Muratona Avenue,
Palini, 15350, Greece

PRODUCT REGISTRATION HOLDER

NVS Regulatory Services Sdn Bhd
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Jalan Tropic 13/50, 40675, Shah Alam, Selangor, Malaysia

LEAFLET REVISION DATE

28 November 2024

code of genePharm